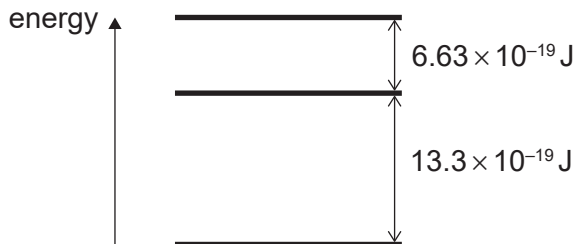


E.1 / PAPER 1 / QUESTION 13

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13. A simple model of an atom has three energy levels. The differences between adjacent energy levels are shown below.



What are the two smallest frequencies in the emission spectrum of this atom?

- A. $0.5 \times 10^{15} \text{ Hz}$ and $1.0 \times 10^{15} \text{ Hz}$
- B. $0.5 \times 10^{15} \text{ Hz}$ and $1.5 \times 10^{15} \text{ Hz}$
- C. $1.0 \times 10^{15} \text{ Hz}$ and $2.0 \times 10^{15} \text{ Hz}$
- D. $1.0 \times 10^{15} \text{ Hz}$ and $3.0 \times 10^{15} \text{ Hz}$